

How Large Language Model Benchmarks Are Built: A Measurement-Oriented Tutorial and Empirical Demonstration with IRT-Viz-Eval

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Abstract

Benchmark construction differs from developing an educational or psychological assessment. This tutorial introduces LLM benchmark development to social-science readers, especially those working in psychological and educational measurement, through eight stages from intended use to maintenance. IRT-Viz-Eval illustrates the process with 84 generated psychometric plots and 198 tasks. Five vision-language systems were administered through Hugging Face Inference Providers and scored against programmatic ground truth. Mean task-normalized scores ranged from .600 to .719; task profiles, output validity, latency, and aggregation differed. The demonstration shows why interpretation requires more than a leaderboard total. Validity and scale-development scholarship disciplines benchmark claims, but benchmark scores are not treated as psychological traits, and the paper does not assume that every benchmark requires a human-test psychometric model.

Keywords: large language models, benchmark design, validity, psychological measurement, item response theory

1 Why Social Scientists Need a Benchmarking Vocabulary

LLMs are increasingly used in research, education, public administration, and professional work. Social scientists may therefore encounter benchmark scores when selecting a model, evaluating an AI-assisted workflow, or making claims about what a model can do. A model may be described as stronger because it scores higher on a broad knowledge benchmark such as MMLU (Hendrycks et al., 2021), a collaborative task collection such as BIG-bench (Srivastava et al., 2023), or a work-sample evaluation such as SWE-bench (Jimenez et al., 2024). Yet a single number can conceal decisions about content, prompts, sampling, scoring, missing responses, and aggregation.

For psychological scientists, the immediate issue is methodological. Benchmark scores may shape model selection, study design, automated coding, and claims about assessment applications even when the benchmark itself has not been examined as a measurement procedure. Understanding how benchmarks are constructed is therefore part of evaluating the evidence behind AI-assisted research and practice.

The social sciences already possess a mature language for asking whether observed performances support intended interpretations. Construct validity, evidence-centered design, reliability, fairness, and consequences are central to educational and psychological assessment (Cronbach and Meehl, 1955; Messick, 1989; American Educational Research Association et al., 2014; Kane, 2013; Mislevy et al., 2003). Practical scale-development guides make the design sequence visible. They define the construct, build and review an overinclusive item pool, pilot the instrument, examine internal structure, and accumulate evidence for intended uses (Clark and Watson, 2019; Boateng et al., 2018; Hinkin, 1998; Simms, 2008). AI

evaluation researchers have begun to apply related ideas to benchmark design. In particular, Evidence-Centered Benchmark Design (ECBD) adapts evidence-centered assessment design to natural language processing and organizes benchmark decisions into capability, content, adaptation, assembly, and evidence modules (Liu et al., 2024). Critical work similarly warns that broad benchmark labels can invite claims that exceed the actual task sample (Raji et al., 2021). This paper builds on that bridge but is written as a practical introduction. It asks what an LLM benchmark is, how one is made, what can go wrong, and what must be reported.

Three contributions distinguish this article from existing benchmark frameworks and audits. First, it translates benchmark development into a step-by-step vocabulary for social scientists who know assessment and validity but may not know model APIs, prompt adaptation, or benchmark maintenance. Second, it connects each design stage to a fully reproducible worked example, including stimuli, task records, response provenance, scoring, and uncertainty. Third, it uses five complete model administrations to demonstrate that routing, invalid outputs, task composition, and aggregation are not implementation details; they change the interpretation supported by a benchmark score. ECBD supplies a modular evidentiary architecture, BetterBench supplies a lifecycle audit, and HELM demonstrates standardized model comparison (Liu et al., 2024; Reuel et al., 2024; Liang et al., 2022). The present article complements those contributions with an assessment-facing tutorial that follows one benchmark from intended use through empirical administration.

Here, a *benchmark* is a standardized procedure for eliciting observable behavior from one or more AI systems and comparing that behavior with a specified reference, rubric, distribution, or set of desiderata. A benchmark commonly includes data, tasks, prompts, scoring code, aggregation rules, model configuration, and documentation. A leaderboard is only one possible presentation of its results. A dataset alone is not a complete benchmark because it does not fully specify how a model encounters the data or how outputs become claims.

IRT-Viz-Eval is used as a running case. It asks whether multimodal LLMs can interpret psychometric plots such as item characteristic curves, item and test information functions, test characteristic curves, category response curves, and non-monotonic ideal-point curves. The case is useful because the underlying mathematics provides exact ground truth, while the visual presentation can be manipulated independently. The benchmark itself remains substantively important, but its primary role in this article is to make the benchmark-construction process concrete.

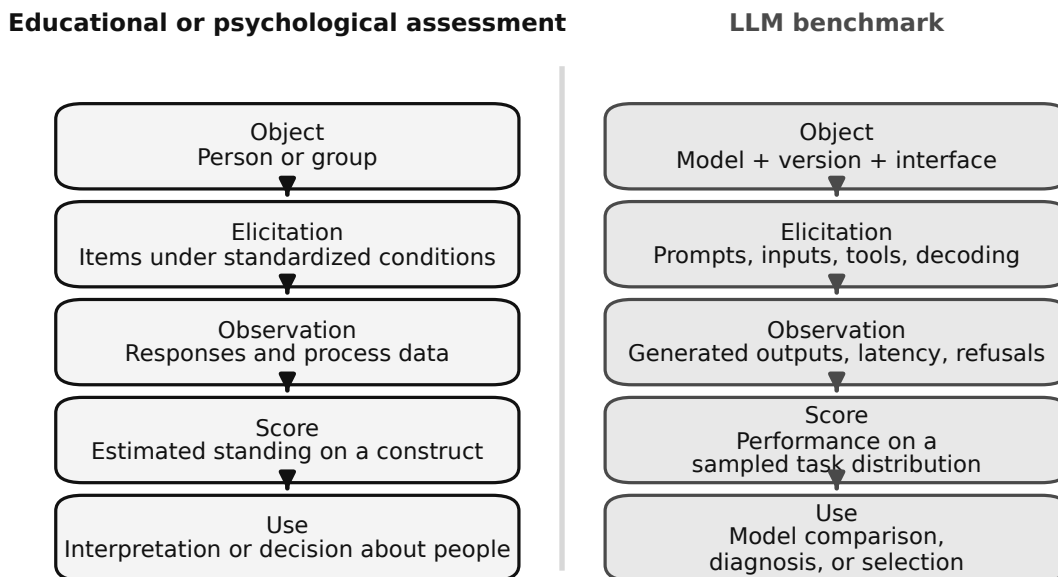
Educational measurement researchers are increasingly encountering AI in automated scoring, feedback, content analysis, and other assessment workflows. Recent educational-measurement scholarship has paired those opportunities with concerns about validity, reliability, fairness, transparency, and responsible implementation (Bulut and Zheng, 2024; Bulut et al., 2024; Huang et al., 2025). Survey evidence also points to a need for stronger connections between machine-learning practice and measurement principles (Zheng et al., 2025). Benchmark development is one part of that methodological problem. Before a score can inform research or practice, the evaluated system, elicitation conditions, scoring rules, and intended interpretation must be explicit. The demonstration compares five model routes from multiple developer and compute-provider ecosystems under a shared interface, retaining the requested model, provider route, configuration, and access date as part of each administration record.

The article is therefore a methodological tutorial with an empirical demonstration, not an attempt to establish a broad model leaderboard. Diagnostic baselines establish that the software behaves as designed; five real model administrations then show how benchmark evidence changes once provider routing, malformed or empty outputs, latency, token limits, and aggregation rules enter the study. The distinction matters. Real-model results are necessary for claims about actual systems, but they do not substitute for construct definition, content review, scoring validation, or a defensible interpretation. A subtle but important boundary governs the paper. Its main object is the development and reporting of an LLM

benchmark, not the psychometric validation of a benchmark as an AI trait measure. Human-assessment concepts are used as design resources and analogies, while claims about latent structure, invariance, or score use remain empirical questions for later studies.

2 A Benchmark Is Test-Like, but It Is Not a Human Assessment

The analogy between a benchmark and a test is productive if its limits are explicit. Figure 1 places the two traditions side by side. In both, an object is placed under elicitation conditions, observable responses are scored, and the scores are used to support an inference. The major differences concern what the object is, how stable it is, what “standardization” requires, and what score use is defensible.



Shared principle: observed performance supports an inference only through an explicit validity argument.

Figure 1: Parallel anatomy of human assessment and LLM benchmarking. The figure is not claiming that the two enterprises are equivalent. It identifies their shared inferential structure while foregrounding the distinct object and conditions of an LLM benchmark.

2.1 The object is a configured system

In most educational and psychological assessments, the object of inference is a person or group. In an LLM benchmark, “the model” is underspecified. Observable behavior may depend on model family, checkpoint or dated version, system message, user prompt, examples supplied in context, image resolution, enabled tools, retrieval corpus, temperature, maximum output length, and provider-side changes. The object is therefore better represented as a configured system comprising a model, version, interface, and adaptation condition. A result without this provenance may not be reproducible even when the benchmark items are public. HELM made standardization and release of raw prompts and completions central to transparent model comparison (Liang et al., 2022); model cards and datasheets likewise emphasize documentation of intended use, provenance, and performance conditions (Mitchell et al., 2019; Gebru et al., 2021).

2.2 The response process is programmable and potentially stochastic

Human response processes vary, but a model adds distinctive sources of variability. A generative model can produce different outputs for the same input because of stochastic decoding or provider implementation. It may also return prose when a label was requested, refuse an item, call a tool, or generate malformed structured data. Prompt wording is part of the administration condition, not a neutral wrapper. Even answer-option order can change accuracy on a familiar multiple-choice benchmark (Gupta et al., 2024). Consequently, a benchmark should preserve raw outputs, define parse-failure handling before analysis, and repeat administrations when stochasticity is relevant.

2.3 Exposure can invalidate the interpretation

In conventional testing, advance access to secure items threatens score interpretation. Public LLM benchmarks face an analogous but harder problem because model training corpora are enormous, often undisclosed, and may contain benchmark questions, answers, code, discussions, or task instructions. Contamination can therefore make memorization look like capability (Sainz et al., 2023). Procedures can sometimes detect exposure in black-box models with formal error control, but the general problem cannot be assumed away (Oren et al., 2024). Programmatic generation, newly sampled forms, private holdouts, release dates, and contamination audits are design responses, not complete guarantees.

2.4 Scores usually describe a task distribution, not a latent trait

A human test score is often interpreted as an estimate of standing on a psychological or educational construct, with a measurement model linking items to persons. Many LLM benchmark scores are simpler empirical summaries, such as average exact-match accuracy, mean rubric points, pass rate, preference rate, or error. They describe performance on sampled tasks under specified conditions. They do not automatically constitute a latent-variable scale, remain invariant across prompt changes, or support claims about general intelligence. The same benchmark name can also conceal a mixture of qualitatively different tasks. Data composition can affect absolute scores and model rankings, sometimes more than the choice of metric (Kovatchev and Lease, 2024). Scores from benchmarks aimed at similar capabilities may fail to concur (Sun et al., 2023).

2.5 The scorer may itself be a model

Closed-form answers can often be scored deterministically. Open-ended generations require exact matching, structured rubrics, expert judgment, crowd judgment, learned metrics, or an LLM-as-a-judge. Model judges can scale evaluation, but they are measurement instruments

and require validation. Agreement with human annotation varies markedly across tasks, properties, and judge models (Bavaresco et al., 2025). A benchmark should therefore distinguish target model from judge model, preserve judge prompts and versions, and report human agreement or adjudication for a defensible sample.

The online supplement provides a detailed comparison of the two design traditions. The implication is neither that AI benchmarks are invalid nor that they must copy human testing. Rather, familiar validity principles need to be translated to a different object. Figure 2 summarizes that translation. Design moves forward from an intended use to capability claims, observable evidence, tasks, and scores. Validation works backward and across the chain, asking whether each link is supported.

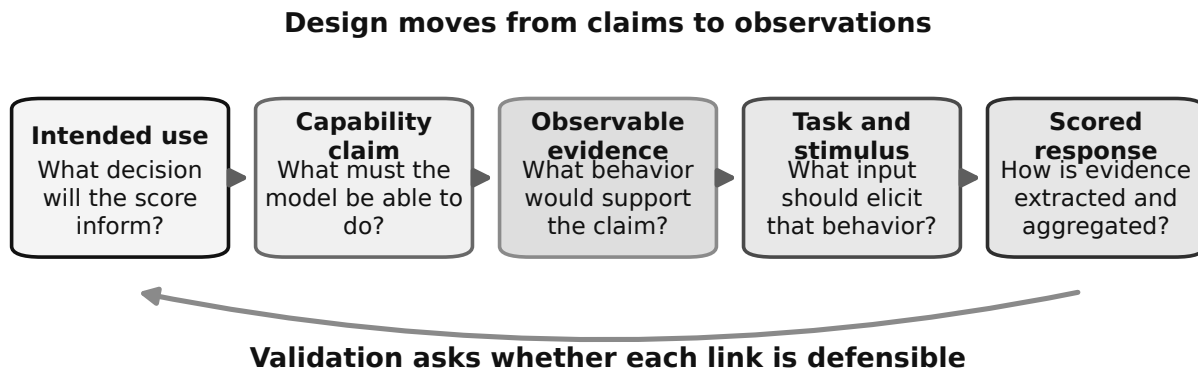


Figure 2: The claim-to-evidence chain. A benchmark item is useful only when its response provides evidence relevant to a stated capability, and that capability is relevant to an intended use. Validation examines every link rather than treating the final score as self-interpreting.

3 How an LLM Benchmark Is Built

Figure 3 presents an eight-stage construction process. It combines the evidentiary logic of ECBD (Liu et al., 2024), the lifecycle perspective of BetterBench (Reuel et al., 2024), and practices from behavioral testing, transparent evaluation, and dataset documentation. The stages are presented sequentially for teaching, although real development is iterative. The

sequence is deliberately recognizable to assessment readers because benchmark construction also requires an explicit content specification, item or task review, pilot administration, scoring decisions, and revision. The analogy stops there. The benchmark object is a configured AI system, and its score usually summarizes performance over a task distribution rather than estimating a person parameter.

An LLM benchmark is a maintained measurement system, not just a question file

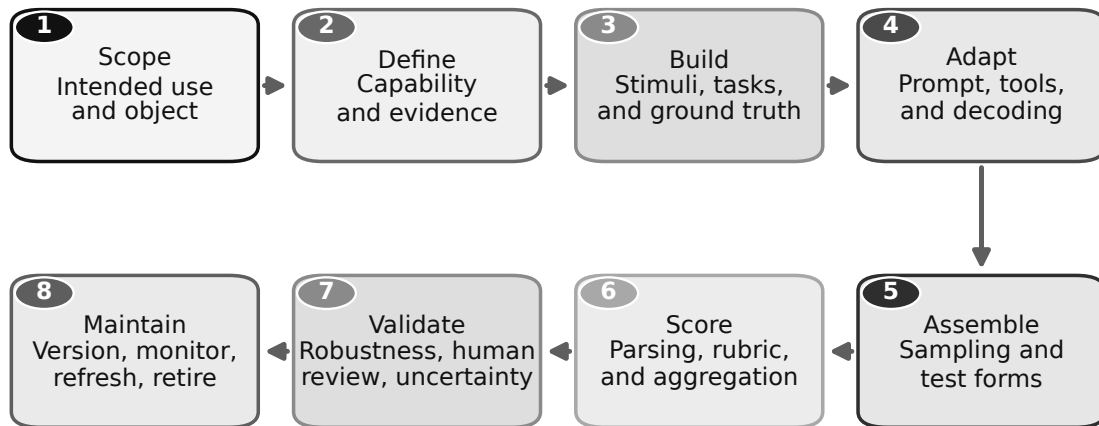


Figure 3: An eight-stage benchmark workflow. Scope and construct decisions precede data generation; scoring is designed before claims are made; validation and maintenance are part of the benchmark rather than optional activities after publication.

3.1 Stage 1: Specify intended use and object of evaluation

Begin with the decision the benchmark is expected to inform. Candidate uses include scientific diagnosis of model behavior, comparison of models under a controlled research protocol, selection for a specific workflow, monitoring after deployment, or testing a hypothesized failure mode. “Measure AI ability” is too broad. A useful statement names users, systems, conditions, and boundaries. For example, “IRT-Viz-Eval supports researchers and psychometric teams comparing vision-capable models for low-stakes assistance in interpreting clean, static IRT

plots. It does not certify unsupervised operational decisions.”

The object of evaluation must be equally explicit. Is it a base model, chat-tuned model, hosted product, retrieval-augmented application, or agent with tools? Can it access the internet? Is the image passed at native resolution? Are prompts zero-shot, few-shot, or optimized separately for each model? A benchmark result applies to this configuration.

3.2 Stage 2: Define capabilities and observable evidence

Capability labels should be narrow enough to operationalize. “Reasoning” is not an item specification. “Identify the lower asymptote of a dichotomous item characteristic curve” or “estimate response probability at a marked trait value” is more useful. For each capability, designers should state what a successful output would look like and what competing process could produce the same answer. This is the benchmark equivalent of distinguishing construct-relevant variance from construct-irrelevant variance.

In IRT-Viz-Eval, visual psychometric interpretation is decomposed into model identification, parameter estimation, probability reading, information-peak localization, category-curve reasoning, test-level interpretation, and non-monotonic peak detection. The evidence is not merely a correct label. Numeric estimates and a brief rationale indicate which visible features the model used. The rationale is supportive process evidence, not privileged access to the model’s internal computation.

3.3 Stage 3: Construct stimuli, tasks, and ground truth

There are three common content sources. First, experts can author and annotate items. This supports ecological relevance but is expensive and can introduce disagreement. Second, existing datasets can be repurposed. This is efficient but may import poorly documented assumptions and high contamination risk. Third, items can be generated programmatically or by models. This can produce scale, controlled variation, and exact answers, but synthetic regularities may allow shortcuts.

The content specification should cross task types with capabilities, domains, difficulty conditions, and relevant populations or contexts. Scale-development guidance likewise recommends beginning with a clear construct definition and an intentionally broad item pool before reducing it through evidence, rather than allowing available items to define the construct after the fact (Clark and Watson, 2019; Hinkin, 1998; Haynes et al., 1995; Sireci, 1998). Behavioral testing is useful here. CheckList distinguishes minimum functionality tests, invariance tests, and directional expectation tests rather than relying only on aggregate held-out accuracy (Ribeiro et al., 2020). IRT-Viz-Eval includes minimum functionality tests such as reading a probability, and invariance tests in which the same curve is rendered under different styles. Figure 4 shows the current content matrix. Empty cells are informative because they reveal where the benchmark does not provide evidence.

The content specification crosses visual objects with observable capabilities

Psychometric visual object	Dichotomous ICC	included	included	included			
	Item information	included			included		
	Polytomous curves	included		included		included	
	Test characteristic	included		included			
	Test information	included			included		
	Ideal-point curve	included		included			included
		Identify model	Estimate parameters	Read probability	Locate information	Reason across categories	Detect non-monotonicity
Capability sampled by the task							

Figure 4: IRT-Viz-Eval content specification. Rows define psychometric visual objects; columns define observable capabilities. Filled cells identify included task combinations. The matrix makes both coverage and omissions visible before a total score is calculated.

Ground truth should be generated or annotated independently of the target model whenever possible. Executable tasks can be checked by running code or tests, as in SWE-bench (Jimenez et al., 2024). Quantitative figures can be linked to their underlying data

table; ChartQA, for example, explicitly studies visual and logical reasoning over chart images and data (Masry et al., 2022). IRT-Viz-Eval stores generating parameters and coordinates alongside each image, allowing exact answers to be derived rather than judged impressionistically.

3.4 Stage 4: Standardize adaptation and elicitation

An LLM rarely consumes a task without adaptation. The benchmark must specify the system prompt, user prompt, examples, output schema, media transformations, tool access, and decoding parameters. Adaptation can be standardized across models for comparability or tailored to each model for best-achievable performance; these answer different questions. A fair study may report both.

Prompts should be piloted for comprehension and unintended cues. In human assessment, response-process evidence asks whether respondents engage the task in the way the interpretation assumes (Padilla and Benitez, 2014). For an LLM, the analogous evidence concerns prompt interpretation, visual access, output-schema compliance, and the distinction between a capability failure and a formatting failure. Structured output simplifies scoring but creates format-following demands. If JSON is required, researchers should report malformed-output rates rather than silently discarding them. When prompts are varied, results should be treated as a distribution over plausible administrations. Changes to answer order, formatting, or labels can function as invariance checks rather than nuisances (Gupta et al., 2024).

3.5 Stage 5: Assemble forms and choose a sampling distribution

Benchmarks often conflate the item pool with the evaluated form. The assembly step determines which items a model actually receives and therefore what the score weights. Sampling should reflect the intended claim. Equal numbers per category estimate a balanced construct profile; prevalence-weighted sampling estimates performance under a target use distribution. Both are legitimate, but they are not interchangeable.

Designers should specify exclusions, duplicate handling, train-development-test separation, random seeds, and whether all models receive identical forms. A benchmark should be large enough to make relevant comparisons stable, but item count alone does not guarantee coverage. BetterBench found that uncertainty reporting and easy replication were uncommon even among established benchmarks (Reuel et al., 2024). Bootstrap intervals over items, repeated runs, and hierarchical models can separate item, model, and administration variation.

3.6 Stage 6: Score, aggregate, and preserve provenance

Scoring begins with the response representation. Exact matching is transparent but brittle. Numeric tolerance bands recognize approximate visual reading. Executable validation is strong when a task has a deterministic success condition. Human rubrics support nuanced outputs but require rater training and agreement analysis. LLM judges reduce cost but need task-specific human validation (Bavaresco et al., 2025).

Aggregation requires substantive justification. A mean can allow strength in one domain to compensate for failure in another. A worst-group or minimum-capability score encodes a different use. Accuracy and efficiency may conflict. HELM’s multi-metric design illustrates why performance, robustness, fairness, calibration, toxicity, and efficiency should not automatically collapse into one rank (Liang et al., 2022). At minimum, report total and disaggregated results, denominators, uncertainty, parse failures, refusals, and cost or latency where relevant.

3.7 Stage 7: Validate the benchmark and its interpretation

Validation is an accumulating argument, not a one-time coefficient. Useful evidence includes expert content review, item-level error analysis, comparison with human judgments, expected ordering of diagnostic baselines, robustness to nuisance transformations, convergence or divergence with related benchmarks, and prediction of performance in the intended workflow. Educational measurement reviews show that validity evidence is often incompletely reported

across tests, especially evidence concerning consequences (Cizek et al., 2008; Downing, 2003; Cook and Beckman, 2006; Lane, 2014). Applied scale research similarly demonstrates why a single internal-consistency coefficient is not a sufficient account of structural validity and why questionable measurement practices should be made visible (Flake et al., 2017; Flake and Fried, 2020; Hussey and Hughes, 2020). If a future project makes trait-like claims, factor-analytic structure and invariance would require their own design and evidence rather than being assumed from a benchmark total (Fabrigar et al., 1999; Reise et al., 2000; Putnick and Bornstein, 2016). Disagreement with another benchmark is not automatically a flaw, but it requires an account of why the constructs or task distributions differ (Sun et al., 2023).

Benchmark quality should itself be tested. Does an oracle obtain the ceiling? Does a deliberately corrupted response receive partial credit? Does a blind response perform near a meaningful floor? Do small visual changes that preserve the answer leave scores stable? Do changes intended to increase difficulty reduce performance? These checks resemble unit tests, simulation studies, and known-groups validity evidence.

3.8 Stage 8: Maintain, refresh, and retire

A public benchmark changes after release even if its files do not. Items can enter model training corpora, developers can optimize against the leaderboard, tasks can saturate, provider versions can drift, and intended uses can change. Dynabench treats data collection and evaluation as a dynamic human-and-model-in-the-loop process (Kiela et al., 2021). A less resource-intensive project can still version releases, retain private or newly generated forms, publish change logs, monitor ceiling effects, and state retirement criteria.

Figure 5 makes this lifecycle explicit. A benchmark has a validity window. During this period, its evidence supports stated comparisons under documented conditions. Release is therefore the start of monitoring, not the endpoint of design.

Publication begins the benchmark lifecycle; it does not end it

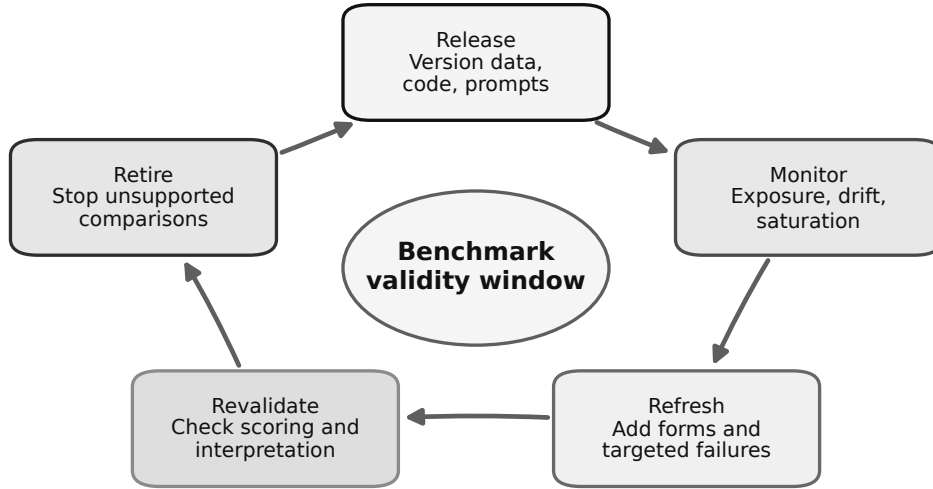


Figure 5: Benchmark maintenance cycle. Monitoring addresses exposure, model drift, and saturation; refreshes require renewed validation; retirement prevents unsupported historical comparisons from being treated as current evidence.

4 Worked Example: Constructing IRT-Viz-Eval

4.1 Intended use and capability model

IRT-Viz-Eval is intended to support research on whether vision-capable language models can interpret visual artifacts used in educational measurement. It is not a certification exam for psychometricians and does not support autonomous high-stakes use. Its present domain is clean, static plots with known generating parameters. The capability claim is correspondingly bounded. Under a documented prompt and image input condition, a model can recover psychometrically relevant properties from an IRT figure and explain the visible evidence for its answer.

The substantive domain includes dichotomous 1PL, 2PL, 3PL, and 4PL item characteristic curves (ICCs); item information functions (IIFs); partial credit, generalized partial credit, and graded response category curves; test characteristic curves (TCCs); test information

functions (TIFs); and a transparent non-monotonic ideal-point family. These objects have visible signatures connected to established IRT parameters (Rasch, 1960; Birnbaum, 1968; Lord, 1980; Samejima, 1969; Andrich, 1978; Muraki, 1992; Roberts et al., 2000).

4.2 Programmatic content and factorial rendering

All stimuli are generated on a shared ability grid from $\theta = -4$ to 4. Dichotomous response probabilities use

$$P_i(\theta) = c_i + \frac{d_i - c_i}{1 + \exp[-1.7a_i(\theta - b_i)]}, \quad (1)$$

where a_i is discrimination, b_i is difficulty, c_i is the lower asymptote, and d_i is the upper asymptote. Item information is

$$I_i(\theta) = \frac{[P'_i(\theta)]^2}{P_i(\theta)[1 - P_i(\theta)]}. \quad (2)$$

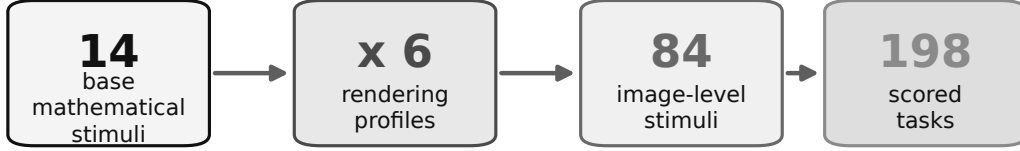
TCCs and TIFs sum item response and information functions. Polytomous curves are produced from PCM, GPCM, and GRM probability functions. The non-monotonic stress test uses

$$P_i(\theta) = l_i + (u_i - l_i) \exp \left[-\frac{1}{2} \left(\frac{\theta - \delta_i}{\sigma_i} \right)^2 \right]. \quad (3)$$

This final family is an interpretable ideal-point curve rather than a full GGUM implementation; its purpose is to test detection and localization of non-monotonicity.

Programmatic generation separates mathematical content from presentation. As Figure 6 shows, the local validation contained 14 base mathematical stimuli. Each was rendered under six style profiles, creating 84 images and 198 linked tasks. This factorial structure makes it possible to compare responses to the same mathematical object under changed visual conditions.

Programmatic generation separates mathematical content from visual presentation



A full local pipeline run then created 594 responses and judgments across three diagnostic baselines.

Figure 6: Expansion of the local IRT-Viz-Eval validation set. The multiplication by rendering profile changes visual presentation while preserving the underlying parameters and expected answers. Different images generate different numbers of tasks, producing 198 total task records.

4.3 Invariance as a designed source of evidence

Six profiles vary background, palette, gridline strength, tick density, legend placement, and line width. They are Matplotlib light, Seaborn white-grid, ggplot-like gray, grayscale without a grid, dark high-contrast, and lattice-like. Figure 7 displays the same 2PL ICC in all six conditions. Because the mathematical curve is held fixed, a drop associated with style is interpretable as a visual robustness failure rather than a harder IRT item. This is a benchmark analogue of an invariance test (Ribeiro et al., 2020).

The current styles are controlled perturbations, not a representative sample of all real psychometric software. Future forms should add confidence bands, empirical noise, dense multi-item overlays, mislabeled or truncated axes, grayscale printing artifacts, and plots exported from operational systems. These additions would extend the claim from clean synthetic figures toward the figures encountered in practice.

4.4 One benchmark instance as an evidence record

Figure 8 shows why an item image alone is not the benchmark. A complete instance links the image to a standardized prompt, requested response schema, expected fields, field-specific

The mathematics is fixed while the rendering changes

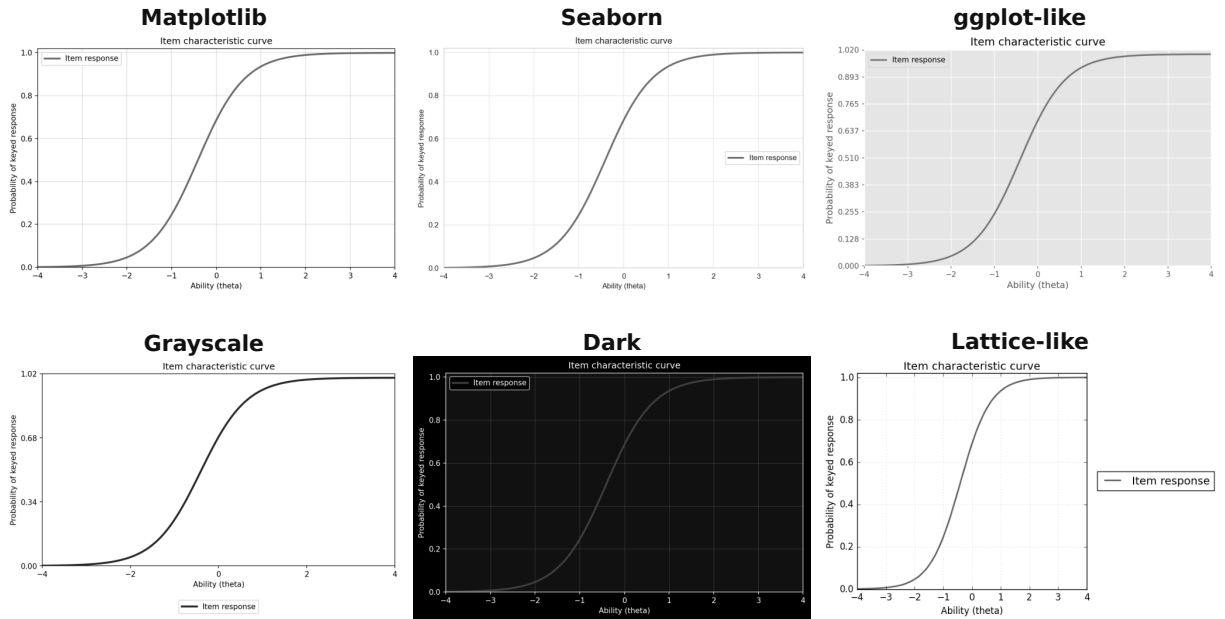
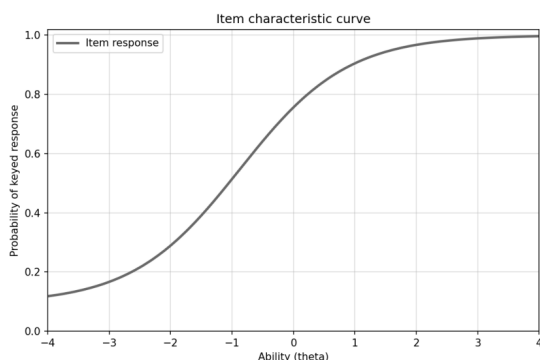


Figure 7: Six renderings of one underlying 2PL item characteristic curve. The curve coordinates and ground truth are identical. Variation in model performance across panels therefore diagnoses sensitivity to presentation rather than psychometric content.

thresholds, provenance, and raw output. The example asks for four dichotomous item parameters. The target output is compact JSON plus a rationale. The benchmark stores both parsed values and the original output so scoring decisions can be audited.

An individual benchmark instance is a complete evidence record

1. Stimulus shown to the model



2. Standardized elicitation

Estimate a, b, c, and d from the item characteristic curve.

Return JSON with the requested fields and a short rationale.

3. Structured observation and scoring

```
{"a": 1.42, "b": -0.36,
  "c": 0.18, "d": 1.00}
```

Compare each field with stored ground truth.
Retain parsing failures, rationale score,
model version, prompt, and latency.

Figure 8: Anatomy of an IRT-Viz-Eval instance. The stimulus, elicitation, structured observation, ground truth, and provenance together form the unit of evidence. Keeping these links prevents an image file or a score from becoming detached from its administration conditions.

The repository writes four linked JSONL files. A stimulus record contains the base stimulus identifier, curve family, IRT model, parameters, coordinate path, image path, and visual metadata. A task record contains the prompt, expected fields, and thresholds. A response record contains model identity, decoding settings, latency, raw output, and parsed output. A judgment record contains field-level scores, rationale score, total, normalized score, and judge provenance. JSON schemas permit automated validation and make missing provenance visible.

4.5 Scoring design

Categorical and integer fields receive exact-match credit after conservative normalization. Numeric fields use task-specific absolute-error bands. In the common parameter example, errors within 0.15, 0.30, and 0.50 receive 3, 2, and 1 points; larger errors receive 0. Probability-reading tasks use tighter bands of 0.05, 0.10, and 0.20. A deterministic rationale rule adds zero to two points based on the presence of prespecified psychometric terms and minimum explanatory content. This component is a reproducible proxy for visible rationale evidence, not a semantic judgment of whether every explanation is correct. Scores are normalized by the maximum available points for that task because task families request different numbers of fields.

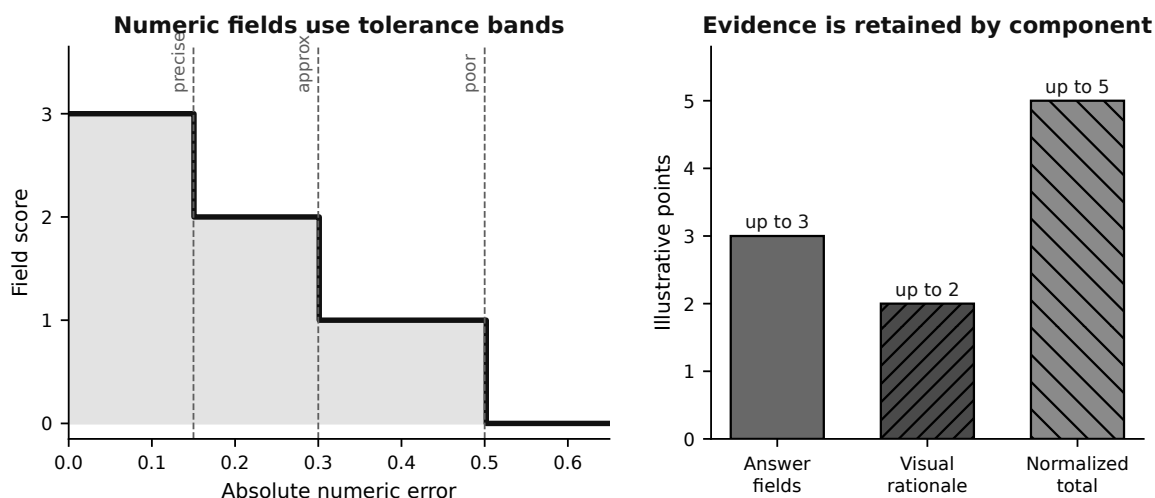


Figure 9: Illustration of IRT-Viz-Eval scoring. The left panel shows tolerance-band scoring for a numeric field. The right panel shows why component scores are retained before normalization. Answer accuracy and rationale evidence can be inspected separately. Point totals are illustrative because the number of answer fields differs by task.

Figure 9 emphasizes two choices. First, visual reading is approximate, so a graded numeric rubric is more informative than exact string equality. Second, the normalized total is not the only result. Field errors, rationale indicators, format compliance, and task type remain available for diagnosis. A future validation study should ask experts to score a stratified response sample and estimate agreement with the deterministic rubric before replacing or

supplementing it with human or LLM judgment.

5 From Benchmark Prototype to Actual Model Administration

5.1 What the local baselines establish

The local validation used seed 20260711 and one sampled base instance per model family. It produced 84 rendered stimuli and 198 tasks. Three deterministic baselines generated 594 responses. An oracle copied expected answers, a noisy baseline perturbed correct answers, and a blind baseline returned generic guesses. These are software diagnostics, not language models and not estimates of real-world performance.

The expected ordering appeared overall and within every task family. The oracle mean normalized score was 1.00, the noisy baseline was 0.73, and the blind baseline was 0.35. The complete diagnostic profile is provided in the online supplement. The result establishes that generation, parsing, field scoring, normalization, and aggregation distinguish exact, approximate, and weak responses. It does not establish content validity, human-model correspondence, or deployed-model competence.

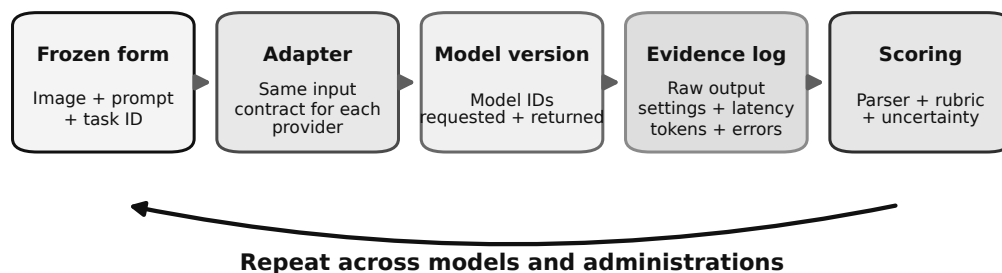
5.2 Choosing an API route

Real-model responses are necessary when a claim concerns an actual system rather than benchmark mechanics. The route used to obtain those responses is part of the administration condition. Direct vendor APIs provide model-native controls; routing services expose models from several developers through one interface; dedicated endpoints provide a controlled hosted deployment; and local inference offers the greatest implementation control when hardware and open weights are available. These routes also differ in transparency, security, cost, and researcher control, concerns documented for generative-model scoring applications

in educational measurement (Ormerod and Kwako, 2024). OpenAI’s Responses API and Google’s Gemini API both accept image inputs, while Hugging Face Inference Providers routes requests to supported third-party models and compute providers through a common client (OpenAI, 2026; Google, 2026; Hugging Face, 2026). The online supplement compares these options and their measurement implications in detail.

Figure 10 shows the common administration path. Each task record supplies an image, prompt, task identifier, output expectation, and scoring rule. A provider adapter sends the same model-visible content to each target system. The resulting evidence record stores the requested and returned model identifiers, access time, output limit, repetition number, latency, token usage, raw output, and any API or parsing error. The scorer operates on these stored records rather than on a live chat transcript.

Administering a benchmark to actual models is a controlled study, not a manual chat



Use diagnostic baselines to test mechanics; use real model calls only when making claims about model performance.

Figure 10: Reproducible administration of IRT-Viz-Eval to actual models. Consumer product labels such as “ChatGPT” are insufficient provenance; a study should record the API model identifier and returned version whenever available. Diagnostic baselines and real model calls answer different questions and should remain separate.

The repository implements both an OpenAI multimodal Responses API adapter and the Hugging Face adapter used in this study. The latter used Hugging Face authentication and routing but explicitly selected the underlying inference provider in each model identifier. This distinction is important. Hugging Face supplied the common interface, model discovery,

authentication, and request routing. Novita served GLM-4.5V and Qwen3-VL 30B, DeepInfra served Gemma 3 4B and Gemma 3 12B, and OVHcloud served Qwen2.5-VL 72B. A consumer subscription to ChatGPT or Gemini is not an API administration record and does not by itself supply API credits. Manual consumer-chat administration was avoided because routing, hidden instructions, media preprocessing, and versioning are difficult to audit.

5.3 Empirical administration protocol

The benchmark was administered on July 11–12, 2026. Five target systems were requested through Hugging Face Inference Providers. Each system received all 198 zero-shot tasks, the same PNG images and prompts, no tools, and a maximum of 1,200 completion tokens. Each prompt requested task-specific JSON plus a brief visual rationale. Unspecified decoding parameters used provider defaults. The runner used resumable JSONL output, bounded request timeouts, transient retries, and one administration per task. Exact requested and returned identifiers, compute providers, commands, and per-run logs are supplied in the online supplement.

No LLM was used as a judge. Categorical fields were scored by normalized exact match, numeric fields by prespecified tolerance bands, and rationale evidence by the fixed rule described earlier. The primary model score was the unweighted mean of the 198 task-normalized scores. To represent uncertainty over sampled mathematical content, 95% intervals were calculated from 10,000 cluster-bootstrap draws over the 14 base stimuli. The paired interval for the model difference resampled the same base-stimulus clusters. These intervals do not represent within-model sampling variability because only one response per task was collected.

5.4 Five-model results

All five systems completed 198 unique tasks without a recorded API error after deduplication. Qwen2.5-VL 72B had the highest mean task-normalized score (.719, 95% cluster-bootstrap

CI [.656, .788]), followed by Qwen3-VL 30B (.704, [.643, .774]). GLM-4.5V (.628, [.536, .727]) and Gemma 3 4B (.629, [.587, .673]) were nearly tied, while Gemma 3 12B scored .600 ([.549, .653]). The paired intervals indicate that the two Qwen systems were not clearly separated, whereas both were above the Gemma and GLM systems in several pairwise comparisons under this task distribution. These are descriptive comparisons from one administration, not stable model traits. Table 1 also shows a consequential aggregation choice. Raw rubric points give more weight to tasks with more requested fields, whereas task normalization gives every task equal weight.

Table 1: Overall scores and administration diagnostics for the complete 198-task runs.

Model	Mean score	95% CI	Points	Empty output	Median latency
Qwen2.5-VL 72B	.719	[.656, .788]	905/1,236	0.0%	3.48 s
Qwen3-VL 30B	.704	[.643, .774]	873/1,236	0.0%	2.25 s
Gemma 3 4B	.629	[.587, .673]	742/1,236	0.0%	2.14 s
GLM-4.5V	.628	[.536, .727]	668/1,236	16.7%	13.79 s
Gemma 3 12B	.600	[.549, .653]	782/1,236	0.0%	3.04 s

Operationally, the routed systems were not equivalent. GLM produced visible text for 165 tasks and parseable JSON for 152; Gemma 3 4B, Gemma 3 12B, and Qwen2.5-VL 72B produced nonempty, parseable JSON for all 198, while Qwen3-VL 30B produced parseable JSON for 194. Median latency ranged from 2.14 seconds for Gemma 3 4B to 13.79 seconds for GLM-4.5V. Provider-reported total token counts ranged from 84,523 for Gemma 3 12B to 342,107 for GLM-4.5V. These values describe the routed systems under this protocol, not immutable properties of the model weights, because providers can differ in image tokenization, reasoning-token accounting, serving hardware, and defaults.

The overall scores concealed substantial profile differences (Figure 11). Qwen2.5-VL 72B was strongest on information peaks (.979), test expected scores (1.000), and non-monotonic peaks (.963); Qwen3-VL 30B was strongest on model identification (.730) and was close to Qwen2.5-VL 72B on information peaks (.950). GLM-4.5V remained strongest among the smaller systems on probability reading (.873) and information peaks (.946), while Gemma 3

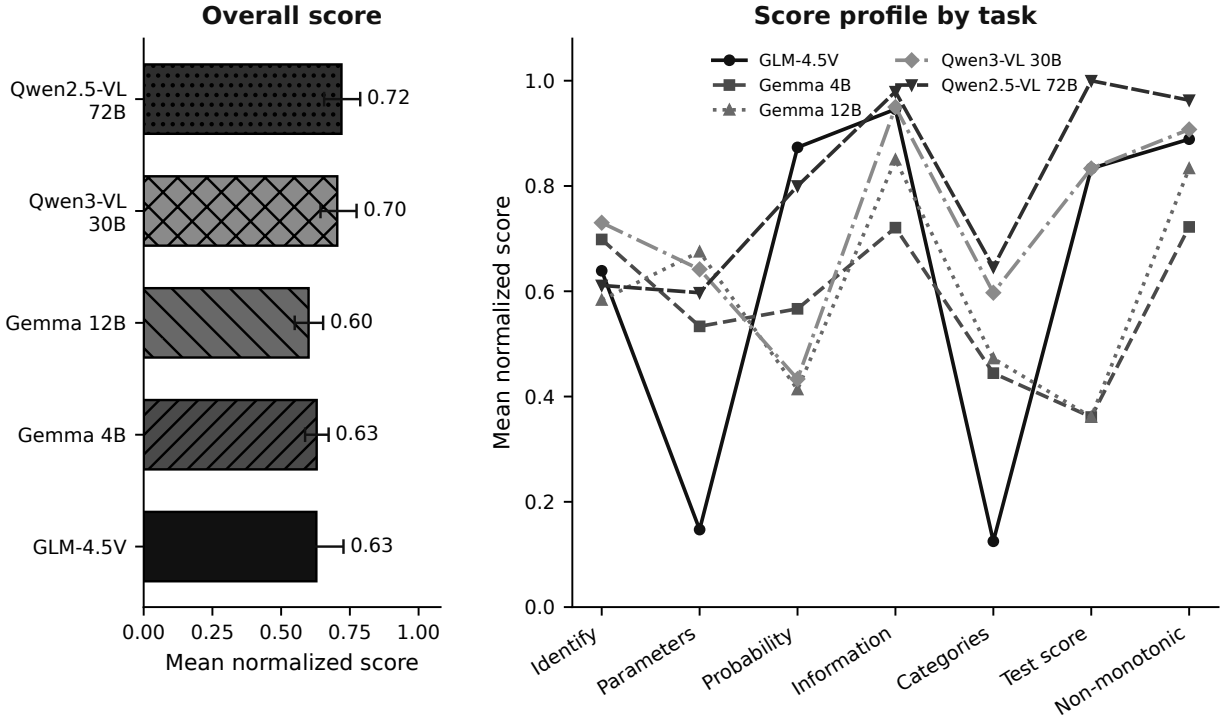


Figure 11: Complete five-model results. Overall bars show mean task-normalized scores with 95% cluster-bootstrap intervals over base mathematical stimuli. Task-level profiles reveal that overall ordering is only one part of the benchmark evidence.

12B led parameter estimation (.675). Gemma 3 4B led category-curve reasoning among the Gemma systems (.444), but the Qwen systems were higher (.597 and .646). The task profile therefore supplies information that a single total suppresses. Exact task-type values are provided in the online supplement. These are descriptive comparisons from one administration; they identify hypotheses for repeated evaluation rather than stable model traits.

Style scores varied within a narrower range than substantive task profiles for all five systems. The ranges were .599–.663 for GLM, .608–.666 for Gemma 4B, .555–.644 for Gemma 12B, .683–.729 for Qwen3-VL 30B, and .687–.746 for Qwen2.5-VL 72B. This descriptive consistency is encouraging evidence that no single plotting theme determined the overall result, but six synthetic styles are not a sample of all operational graphics. Curve-family profiles differed more sharply, especially for test-characteristic and unfolding tasks. The online supplement provides the complete rendering-profile and curve-family figure. This is why style robustness should be reported alongside, rather than substituted for, content coverage.

6 Validity Threats in LLM Benchmarking

Figure 12 locates five threats along the path from stimulus to claim. The diagram is deliberately causal in spirit. An observed score can be altered by factors introduced before, during, or after model generation.

6.1 Contamination and shortcut learning

Prior exposure can occur at the level of task instructions, raw text or images, labels, code, or public discussions (Sainz et al., 2023). Programmatic generation reduces exact-item exposure but does not prevent a model from learning the generating template. Designers should therefore vary parameters and rendering conditions, keep some generators or forms private where appropriate, record release dates, and avoid claims that require proving complete non-exposure.

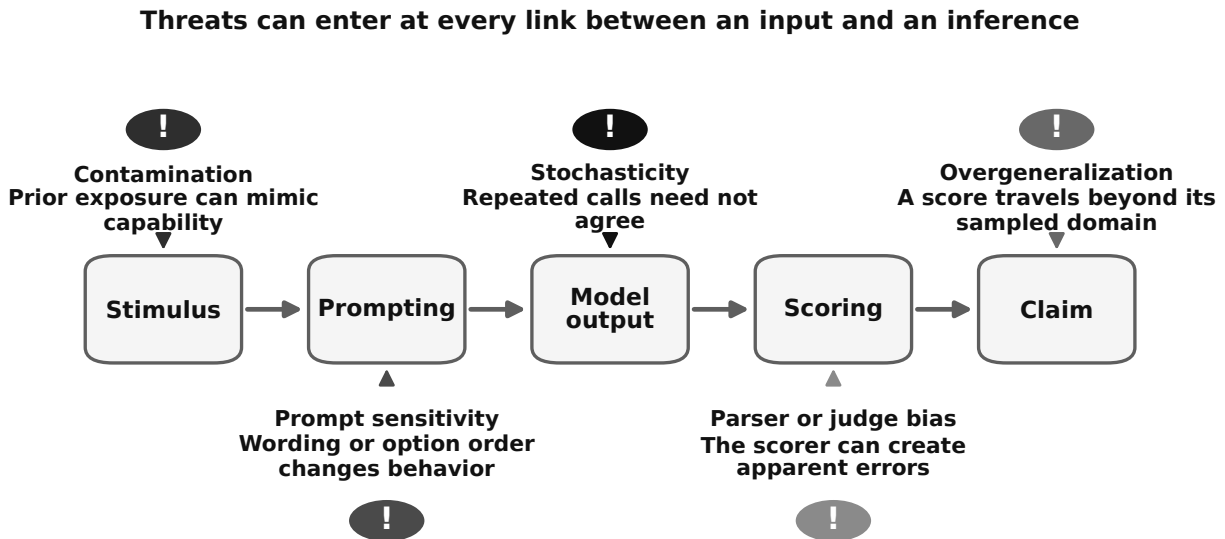


Figure 12: Threats to benchmark inference. Contamination affects the relation between stimulus and capability; prompt sensitivity affects elicitation; stochasticity affects observed output; parser or judge bias affects scoring; overgeneralization affects the interpretation attached to the score.

Shortcuts can arise even without contamination. A model may infer the answer from file-names, alt text, legend labels, repeated color mappings, or template regularities. Benchmark packaging should remove answer-bearing metadata from the model-visible input. Counterbalanced and adversarial variants can test whether a purported capability survives removal of the cue.

6.2 Prompt and interface sensitivity

The prompt determines what behavior is elicited. A model can fail because it lacks the target capability, misunderstands an instruction, cannot emit the requested format, or follows a conflicting system policy. These are different findings. Reporting multiple plausible prompts and invalid-output rates helps distinguish them. A benchmark intended to compare model products should also treat provider preprocessing and tool access as part of the product, while a benchmark intended to compare base capabilities may disable them.

6.3 Stochasticity and version drift

A single API response is one draw from a configured system. Repeated runs reveal whether model rankings are stable or depend on sampling. Hosted systems may change without a new public name. Access dates, model identifiers, raw outputs, and periodic anchor-item reruns are therefore essential. Historical scores should not be pooled across silent model versions as if they were measurements of one fixed object.

6.4 Scoring and judge error

Parsers can mark semantically correct outputs wrong, while permissive extraction can reward extraneous or contradictory text. LLM judges may exhibit task-specific disagreement with people and each other (Bavaresco et al., 2025). Deterministic scoring should be preferred when ground truth is structured. When judgment is necessary, designers should publish the rubric, blind or randomize comparison order, estimate agreement with relevant human experts, retain individual judgments, and specify adjudication.

6.5 Overgeneralization and consequential validity

Benchmark results often travel farther than their design. A score on synthetic IRT plots does not establish competence with empirical calibration reports, psychometric decision making, or the consequences of wrong advice. Raji and colleagues’ critique of “general” benchmarks is a reminder that dataset breadth is not a license for unbounded claims (Raji et al., 2021); consequences evidence similarly asks what follows when scores are used for decisions (Lane, 2014). The defensible interpretation should name the sampled content, model configuration, score rule, date, and intended use. Any model-selection decision should add use-specific validation rather than rely on a broad leaderboard alone.

7 A Reusable Recipe for Social Science Benchmark Projects

For a social science team beginning an LLM evaluation, the following sequence is a practical minimum.

1. Write one paragraph naming the decision, users, model system, context, and claims that are out of scope.
2. Create a construct map with narrow capabilities and observable behaviors; have subject-matter experts review it.
3. Build a content matrix crossing capabilities with domains, task types, difficulty, and nuisance transformations.
4. Choose an item source and ground-truth process; retain provenance and separate target-model generation from answer creation.
5. Pilot prompts and response schemas on several model families; freeze administration rules before comparison.
6. Define form assembly and analysis denominators, including invalid outputs and refusals.
7. Implement scoring with unit tests, diagnostic baselines, and human validation for judgment-based components.
8. Report disaggregated results and uncertainty; inspect errors before ranking models.
9. Publish documentation, raw outputs where permitted, code, version identifiers, known limitations, and maintenance plans.
10. Refresh, revalidate, or retire when exposure, saturation, model drift, or new intended uses undermine the original argument.

This recipe is intentionally closer to assessment design than to downloading a test set and calculating accuracy. It also remains different from human psychometrics. A team need not fit an IRT model to every benchmark or treat models as people. The transferable insight is evidentiary discipline. Define the claim, design observations that bear on it, standardize

the elicitation, quantify uncertainty, and constrain the interpretation.

8 Discussion

LLM benchmarking and social science measurement have much to offer one another. Benchmark developers contribute practices for executable evaluation, large-scale automation, adversarial testing, and rapid iteration. Social scientists contribute theories of constructs, response processes, validity arguments, fairness, sampling, and consequences. Scale-development research adds a practical discipline of specifying the content domain, reviewing candidate indicators, piloting administration, examining structure, and revising before broad use (Boateng et al., 2018; Simms, 2008). ECBD demonstrates that this exchange is already possible (Liu et al., 2024). BetterBench’s lifecycle analysis similarly shows that benchmark quality depends on design, implementation, documentation, maintenance, and eventual retirement rather than predictive performance alone (Reuel et al., 2024).

IRT-Viz-Eval illustrates the benefits of this combined perspective. Programmatic generation provides exact answers; factorial rendering supports controlled robustness evidence; structured records preserve provenance; field-level scoring supports error analysis; and diagnostic baselines test mechanics. The five real-model administrations add a second lesson. Qwen2.5-VL 72B and Qwen3-VL 30B ranked above the other systems on the aggregate task distribution, but the task profiles crossed repeatedly; the smallest Gemma system was nearly tied with GLM overall; GLM had a 16.7% empty-output rate; median latency differed by more than sixfold; and raw-point and task-normalized aggregation answer different questions. Benchmark construction decisions were therefore not background details; they changed what a comparison would mean.

The validity argument remains bounded. The benchmark samples clean synthetic plots, five routed systems and one response per task were observed, provider defaults were not fully harmonized, the rationale rule needs expert validation, and the six style profiles do not

represent the ecology of operational psychometric graphics. The cluster-bootstrap intervals reflect variation across 14 mathematical base stimuli, not model stochasticity, future provider drift, or uncertainty in scoring. The benchmark has not been developed as a validated latent-trait scale for AI systems, and these empirical results do not establish that any model is suitable for unsupervised psychometric decisions. The educational and psychological measurement literature is therefore used here to improve benchmark construction and score interpretation, not to imply that the present benchmark has completed the evidence needed for psychometric claims about model ability.

The next validation phase should repeat administrations, counterbalance plausible prompts, and obtain blinded expert scores for a stratified response sample. Content extensions should include empirical estimation noise, confidence bands, differential item functioning plots, item-fit diagnostics, score-report visualizations, and accessibility conditions. An external study could then test whether benchmark performance predicts quality on authentic psychometric review tasks.

9 Conclusion

An LLM benchmark is best understood as a maintained measurement system. Its data are only one component. The prompts define elicitation, the model configuration defines the object, the parser or judge defines the observation, the assembly rule defines the sampled domain, and the validity argument defines what the score can mean. These features distinguish benchmarking from conventional educational and psychological assessment while preserving a shared commitment to evidence-based interpretation.

For social scientists, the practical lesson is straightforward. Do not begin or end with a leaderboard. Begin with an intended use and a defensible capability claim. Build backward to the evidence and tasks, then move forward through standardized administration, scoring, validation, reporting, and maintenance. The empirical demonstration showed why. Five

systems differed in overall score, task profile, output validity, latency, and aggregation behavior. IRT-Viz-Eval provides one complete implementation of this benchmark-development process and a bounded foundation for studying multimodal interpretation of psychometric graphics. Its purpose is to teach how such a benchmark is built and administered, not to declare that the benchmark itself is a finished psychometric scale for AI.

10 Declarations

Data and code availability. Code and benchmark materials are available at <https://github.com/pbotter42/IRT-Viz-Eval>. The exact response records, judgments, analysis outputs, and figure-generation materials used for this preprint are included in the accompanying reproducibility archive.

Ethics statement. Ethics review was not required because the study used synthetic benchmark stimuli and model-generated responses and involved no human participants or personal data.

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Online Supplement

How Large Language Model Benchmarks Are Built

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1 Purpose and scope

This supplement documents the benchmark-development decisions, empirical administration records, scoring rules, additional results, and reproducibility materials for the accompanying manuscript. It is intentionally more operational than the main article. The manuscript’s purpose is to explain how an LLM benchmark is built and administered for a social-science audience. This supplement should not be read as evidence that IRT-Viz-Eval is already a validated psychometric scale for AI systems. It supplies the materials needed to inspect, reproduce, extend, and eventually validate the benchmark.

The benchmark is an instrumented evaluation procedure: a frozen task form, an elicitation contract, a configured model route, a response record, a deterministic scorer, and an aggregation rule. Its empirical scores describe these model-route-task configurations. They are not estimates of a human-like latent ability parameter, and the present study does not claim measurement invariance, generalizability across providers, or suitability for unsupervised psychometric work.

2 Human assessment and LLM benchmark design

Table S1: Practical differences between assessments of people and benchmarks of LLM systems.

Design issue	Educational or psychological assessment	LLM benchmark
Object	Person, group, or population	Versioned model system under a specified interface
Administration	Instructions, timing, setting, accommodations	Prompt, context examples, media preprocessing, tools, decoding, retries
Response	Selected or constructed human performance	Generated tokens, tool calls, refusals, latency, malformed outputs
Exposure threat	Item security, coaching, practice	Pretraining, fine-tuning, evaluation leakage, public prompt reuse
Scoring	Key, rubric, rater, measurement model	Parser, executable check, metric, human rater, or model judge
Repeatability	Affected by persons and occasions	Also affected by sampling seeds and silent model updates
Typical claim	Standing on a construct or expected performance	Performance on a sampled task distribution under tested conditions
Lifecycle	Forms are revised and revalidated	Public tasks can saturate or contaminate quickly; versions may drift

3 Study and release manifest

The frozen release contains 14 base mathematical stimuli, six rendering profiles, 84 image-level stimuli, and 198 scored tasks. The number of tasks varies by image because different curve families support different observable questions. Every model received the same 198 task identifiers and native PNG images. The benchmark generator, task records, ground-truth values, image files, scoring code, schemas, prompts, and analysis scripts are included in the accompanying reproducibility archive and at <https://github.com/pbotter42/IRT-Viz-Eval>.

Table S2: Core study objects and their roles.

Object	Role in benchmark development
Base stimulus	Mathematical object with stored parameters, curve family, and exact ground truth.
Rendering profile	Meaning-preserving visual transformation used to probe presentation robustness.
Image-level stimulus	One base stimulus rendered under one profile and linked to an image file.
Task	One elicitation and scoring contract attached to an image-level stimulus.
Response	One model output under a named route, with raw output, latency, tokens, and provenance.
Judgment	Deterministic field-level scoring record retaining errors, rationale points, totals, and normalization.

4 Model administration and provenance

All five complete runs used Hugging Face Inference Providers as the common API route. The compute provider was recorded in the requested model string. This improves interface comparability while preserving provider differences as part of the observed system. A direct OpenAI or Gemini adapter can be used in future work, but a consumer ChatGPT or Gemini subscription is not itself a reproducible API administration condition.

4.1 API and deployment options

Table S3: API and deployment options for administering a multimodal benchmark.

Route	Examples	Primary advantage	Measurement concern
Direct vendor API	OpenAI Responses; Google Gemini	Native model controls and provider metadata	Separate credentials, billing, schemas, and version policies
Model router or hub	Hugging Face Inference Providers	One task adapter can reach several model families	Routing and underlying compute provider must be recorded
Dedicated endpoint	Managed cloud or Hugging Face endpoint	Stable deployment and configurable resources	Setup cost and endpoint configuration become part of the system
Local inference	Transformers, vLLM, or similar runtime	Control over weights, decoding, and logs	Hardware, quantization, runtime, and preprocessing affect comparability

Table S4: Overall results for all complete administrations.

Model	n	Mean	95% CI	Points	Empty	Median latency
Qwen2.5-VL 72B	198	.719	[.656, .788]	905/1236	0.0%	3.48 s
Qwen3-VL 30B	198	.704	[.643, .774]	873/1236	0.0%	2.25 s
GLM-4.5V	198	.628	[.536, .727]	668/1236	16.7%	13.79 s
Gemma 3 4B	198	.629	[.587, .673]	742/1236	0.0%	2.14 s
Gemma 3 12B	198	.600	[.549, .653]	782/1236	0.0%	3.04 s

Table S5: Administration and output-format diagnostics.

Model	Requested route	Returned identifier	Nonempty	Parseable	Tokens
Qwen2.5-VL 72B	Qwen/Qwen2.5-VL-72B-Instruct:ovhcloud	Qwen2.5-VL-72B-Instruct	198/198	198/198	219,565
Qwen3-VL 30B	Qwen/Qwen3-VL-30B-A3B-Instruct:novita	qwen/qwen3-vl-30b-a3b-instruct	198/198	194/198	189,730
GLM-4.5V	zai-org/GLM-4.5V:novita	zai-org/glm-4.5v	165/198	152/198	342,107
Gemma 3 4B	google/gemma-3-4b-it:deepinfra	google/gemma-3-4b-it	198/198	198/198	85,531
Gemma 3 12B	google/gemma-3-12b-it:deepinfra	google/gemma-3-12b-it	198/198	198/198	84,523

Table S6: Task-type results for every complete model run.

Task type	n	Qwen2.5-VL 72B	Qwen3-VL 30B	GLM-4.5V	Gemma 3 4B	Gemma 3 12B
Model identification	84	.611	.730	.639	.698	.583
Parameter estimation	24	.597	.642	.147	.533	.675
Probability at theta=0	30	.800	.433	.873	.567	.413
Information peak	30	.979	.950	.946	.721	.850
Category-curve reasoning	18	.646	.597	.125	.444	.472
Test expected score	6	1.000	.833	.833	.361	.361
Non-monotonic peak	6	.963	.907	.889	.722	.833

Table S7: Curve-family results for every complete model run.

Curve family	n	Qwen2.5-VL 72B	Qwen3-VL 30B	GLM-4.5V	Gemma 3 4B	Gemma 3 12B
Dichotomous ICC	72	.619	.602	.506	.603	.552
Item information	48	.867	.849	.841	.740	.732
Polytomous curves	36	.656	.688	.424	.546	.551
Test characteristic	12	.722	.750	.694	.514	.403
Test information	12	.813	.813	.750	.677	.698
Unfolding	18	.788	.658	.833	.652	.604

Table S8: Rendering-profile results for every complete model run.

Rendering profile	Qwen2.5-VL 72B	Qwen3-VL 30B	GLM-4.5V	Gemma 3 4B	Gemma 3 12B
Matplotlib light	.709	.729	.599	.666	.593
Seaborn white-grid	.721	.709	.646	.643	.584
ggplot gray	.746	.700	.630	.608	.627
Grayscale	.717	.692	.611	.627	.597
Dark high-contrast	.736	.683	.619	.611	.555
Lattice-like	.687	.712	.663	.621	.644

Table S9: All paired cluster-bootstrap differences (row model minus column model).

Model A	Model B	Difference	95% CI
GLM-4.5V	Gemma 3 4B	-0.001	[-0.078, .076]
GLM-4.5V	Gemma 3 12B	.028	[-0.054, .113]
GLM-4.5V	Qwen3-VL 30B	-0.076	[-0.142, -0.008]
GLM-4.5V	Qwen2.5-VL 72B	-0.091	[-0.145, -0.039]
Gemma 3 4B	Gemma 3 12B	.029	[-0.004, .056]
Gemma 3 4B	Qwen3-VL 30B	-0.075	[-0.141, -0.021]
Gemma 3 4B	Qwen2.5-VL 72B	-0.090	[-0.145, -0.041]
Gemma 3 12B	Qwen3-VL 30B	-0.104	[-0.172, -0.047]
Gemma 3 12B	Qwen2.5-VL 72B	-0.120	[-0.179, -0.066]
Qwen3-VL 30B	Qwen2.5-VL 72B	-0.015	[-0.042, .008]

The first GLM response file contained 199 raw rows because one response identifier was duplicated during resumable collection. The preparation script retained one completed record per response identifier, removed the duplicate, and required exactly one unique record for each of the 198 task identifiers. All other complete files contained 198 raw rows and 198 unique tasks. No API-error rows remained in the complete files used for scoring.

5 Diagnostic baseline profiles

The oracle, noisy, and blind baselines test whether the benchmark pipeline distinguishes exact, approximate, and weak responses. They do not estimate model performance or establish content validity. Figure S1 displays overall and task-family separation.

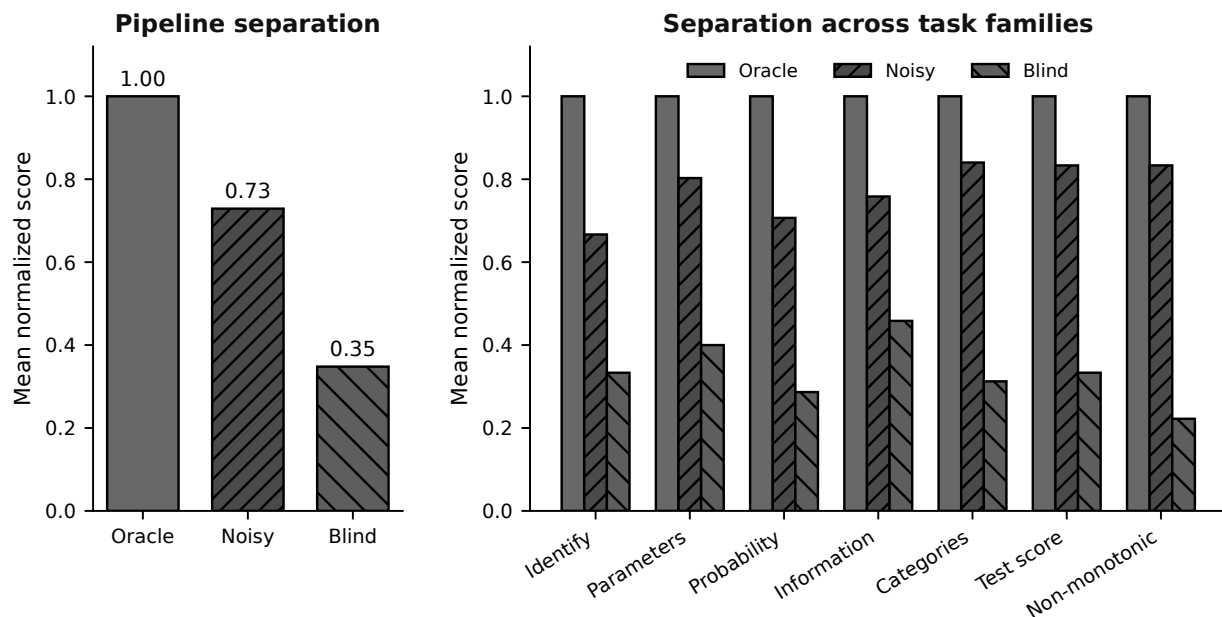


Figure S1: Diagnostic pipeline validation. The left panel shows overall separation of the oracle, noisy, and blind baselines; the right panel shows separation by task family. Because these are constructed local baselines, the graph validates benchmark mechanics only.

6 Rendering and curve-family profiles

Figure S2 disaggregates scores by rendering profile and curve family. The style panels probe controlled presentation changes, whereas the curve-family panels describe substantive content differences.

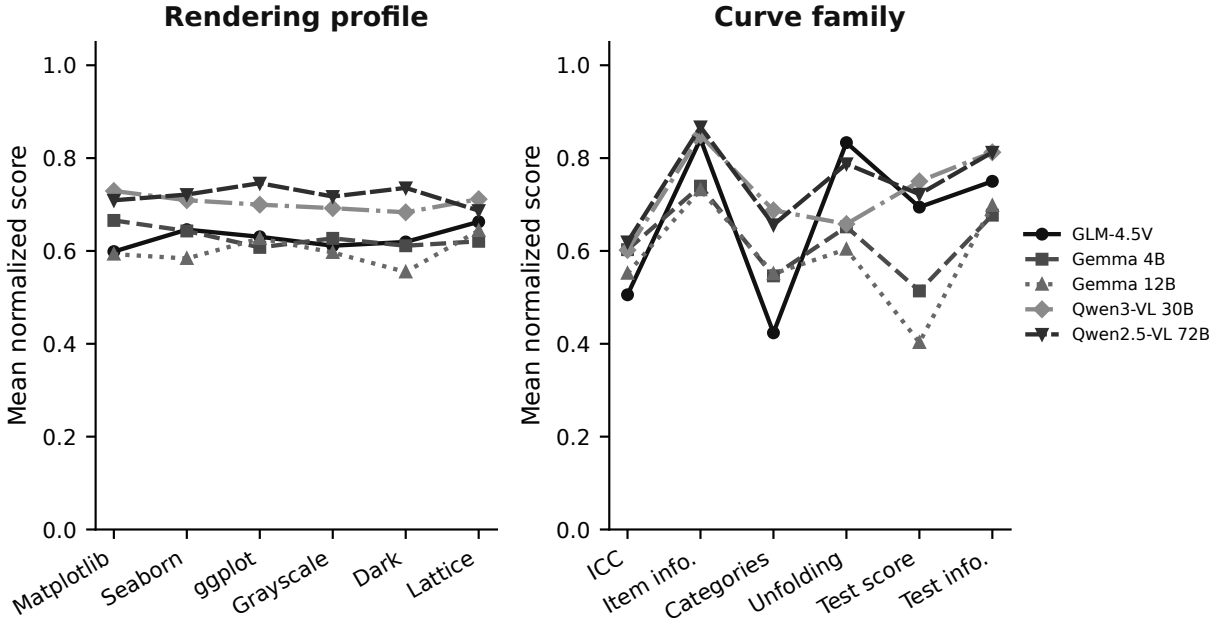


Figure S2: Mean task-normalized scores by rendering profile and curve family. Similar style ranges coexist with larger content-profile differences, illustrating why both nuisance-condition and substantive-domain disaggregation are needed.

7 Frozen elicitation contract

The task prompt is generated from the task record. It supplies the model with one PNG image and a task-specific request. The prompt asks for JSON containing the expected fields and a short visual rationale. The task record, rather than an undocumented conversational instruction, is the source of the requested fields and scoring thresholds.

The model-visible administration contract is:

1. one native benchmark PNG image;
2. one zero-shot task prompt with no demonstrations;
3. no external tools, retrieval, or browsing;
4. a maximum completion length of 1,200 tokens;
5. one response per task in the reported administration;
6. raw output retained before parsing;
7. task-specific JSON fields plus a short rationale;

8. deterministic scoring after collection, with no LLM-as-a-judge.

Provider defaults were not fully harmonized. Temperature, image-token accounting, reasoning-token accounting, hidden serving details, and throughput can differ. The correct interpretation is therefore performance of each named model route under the frozen task and prompt contract, not a claim that model weights alone caused every observed difference.

8 Scoring and missing-output rules

Categorical and integer fields receive exact-match credit after conservative normalization. Numeric fields receive graded credit under task-specific absolute-error bands. For common parameter fields, errors within 0.15, 0.30, and 0.50 receive 3, 2, and 1 points; larger errors receive 0. Probability fields use tighter bands of 0.05, 0.10, and 0.20. The rationale component awards zero to two points using a fixed rule based on prespecified psychometric terms and minimum explanatory content. It is a reproducible proxy for visible rationale evidence, not a semantic judgment of whether a model’s complete explanation is correct.

For each task, the total is divided by the maximum available points for that task. The primary model score is the unweighted mean of these 198 task-normalized scores. This makes each task contribute equally even though tasks request different numbers of fields. Raw point totals are retained as a sensitivity analysis because they weight multi-field tasks more heavily.

Empty outputs, API-error outputs, malformed JSON, missing requested fields, and unparsable values remain in the response and judgment files. A missing or invalid field receives zero for that field; it is not silently removed from the denominator. The main paper reports empty-output and parseable-output rates separately because they describe operational behavior that a score alone can hide.

9 Uncertainty and analysis decisions

The primary uncertainty interval is a 95% cluster bootstrap over the 14 base stimuli. For each draw, base-stimulus clusters are sampled with replacement and all linked tasks within each sampled cluster are retained. This accounts for the fact that six rendering profiles and multiple task records are not independent replications of mathematical content. Pairwise intervals resample the same clusters after aligning model judgments by task identifier.

The cluster bootstrap does not represent within-model sampling variability, because the study collected one response per task. It also does not represent provider drift, future model updates, judge uncertainty, or uncertainty in the deterministic rubric. These are separate validation targets. The supplement therefore reports descriptive intervals without converting them into general claims about stable model rankings.

10 Additional interpretation guidance

The results can support several bounded statements: the named routes produced different levels of performance on this task distribution; Qwen2.5-VL 72B and Qwen3-VL 30B had the highest observed aggregate scores; task profiles differed; and output validity and latency differed materially. The results cannot support the statements that one model has a general intelligence level, that the benchmark is invariant across providers, that success on synthetic plots transfers to operational psychometric practice, or that a model is safe for unsupervised assessment decisions.

This distinction mirrors a central lesson from educational and psychological measurement: scores acquire meaning through design, evidence, and intended use. In this project, that lesson is used to improve benchmark construction. The paper is not claiming that a benchmark total becomes a validated AI psychometric scale merely because it has an item matrix, a rubric, or an IRT-related content domain.

11 Recommended expert-validation study

The next study should obtain independent content and response-process evidence. A practical design is:

1. recruit psychometricians or advanced measurement students to review the capability map and content matrix;
2. ask experts to classify a stratified sample of images by curve family, visual feature, and intended task;
3. collect blinded expert scores for model responses, including partial credit and rationale quality;
4. estimate agreement between the deterministic rubric and expert scoring at the field level;
5. repeat a subset of model calls under the same prompt to estimate within-route stochasticity;
6. counterbalance prompt wording, answer-field order, and output format as explicit administration facets;
7. add empirical plots with estimation noise, confidence bands, dense overlays, truncated axes, and accessibility constraints;
8. compare benchmark performance with a separate authentic psychometric-review task, without treating that correlation as automatic proof of construct validity.

This design would add evidence about content representation, response processes, relations to an external task, and consequences. It would still require a separate argument before the score could be used for operational decisions.

12 Repository map and reproduction

The following files are the core reproducibility objects:

Table S10: Repository map and contents.

Path	Contents
<code>data/benchmark/stimuli.jsonl</code>	Stimulus records, parameters, ground truth, and visual meta-data.
<code>data/benchmark/tasks.jsonl</code>	Task prompts, expected fields, thresholds, and image links.
<code>data/benchmark/images/</code>	PNG files shown to models.
<code>data/empirical/</code>	Deduplicated model response records used for analysis.
<code>paper/prepare_empirical_results.py</code>	Deduplication, scoring, uncertainty, and result-table preparation.
<code>paper/generate_figures.py</code>	Reproducible generation of all main-paper figures.
<code>paper/generate_supplement.py</code>	Reproducible generation of supplemental empirical tables.
<code>tests/</code>	Unit tests for generation, schemas, scoring, and provider behavior.

After dependencies are installed, the empirical tables and figures can be regenerated with:

```
PYTHONPATH=src python3 \  
  paper/prepare_empirical_results.py  
PYTHONPATH=src python3 \  
  paper/generate_supplement.py  
PYTHONPATH=src python3 \  
  paper/generate_figures.py --grayscale \  
  --output-dir paper/figures_grayscale
```

A new provider administration should first use a small pilot, preserve the exact requested model and provider string, and write to a new JSONL path. A full run should use the same output path when resumed so successful response identifiers can be skipped. API keys must be supplied through the local environment or Hugging Face credential store and must never be committed.

13 Supplemental reporting checklist

Before releasing a new benchmark version, record: the intended use; the exact model object; capability definitions; content and exclusion rules; task and image generators; ground-truth derivation; prompt and output schema; provider and access date; timeout and retry policy; raw output retention; parser and scorer versions; missing-output handling; uncertainty procedure; human or model-judge provenance if applicable; contamination risks; licenses; version changes; maintenance owner; and unsupported interpretations.